J.A.R.V.I.S. Android Voice Assistant

Version 2.0 System Architecture, New Features & Complete Command Reference

Target Device: iQOO 12 (Android 14 / Funtouch OS 14)	**Release Version:** Version 2.0 (v2)
Architecture: 100% On-Device / Zero Cloud Dependency	**Voice Profile:** Natural British Male (en-gb-x-rjs)

1. What's New in Version 2.0 (v2)

Version 2.0 upgrades JARVIS from a reactive voice utility into an intelligent, contextual on-device assistant. Every feature runs strictly locally without battery strain or cloud subscriptions.

- **Spoken Notification & Message Reader:** Integrates JarvisNotificationListenerService to capture incoming notifications. Users can ask *'Read notifications'* or *'Read WhatsApp messages'* to hear unread alerts hands-free.
- **Context Memory & Relationship Mapping:** Powered by MemoryRepository. JARVIS remembers the user's name (*'My name is Sarat'*) and relationship mappings (*'Remember Teja is my brother'*). Calling *'Call my brother'* automatically resolves and calls the assigned contact.
- **Expanded Accessibility Controls:** Instant voice commands for *'Take a screenshot'* (via GLOBAL_ACTION_TAKE_SCREENSHOT), *'Open notifications'*, *'Quick settings'*, *'Power menu'*, and *'Split screen'*.
- **Deep Media & Music Automation:** Direct in-app voice playback for Spotify (*'Play Believer on Spotify'*) and YouTube (*'Play Interstellar on YouTube'*).
- **Tactical System Diagnostics & Personality:** Requesting *'Status report'* or *'System diagnostics'* triggers an authentic Marvel-style tactical readout detailing battery percentage, charging state, accessibility status, and notification listener readiness.
- **Focus Mode Routine:** Saying *'Focus mode'* or *'Work mode'* silences ringers, enables Do Not Disturb, and disables flashlight.
- **Pre-Cached Performance:** All launcher metadata and contact books are cached in memory for sub-millisecond execution times.

2. System Architecture & Core Modules

Module / Component	Responsibility & Implementation Details
JarvisAssistActivity	Default Digital Assistant intent target. Holding the device Power button launches the assistant completely silently with a micro-haptic vibration cue.
JarvisOverlayService	Renders the compact Siri-style bottom floating pill HUD (~120dp height) with real-time fluid neon sine wave animation responding to microphone rmsLevel.
VoiceListeningService	Continuous background wake-word listening service for 'Hey Jarvis' with auto-rebind lifecycle resilience.
JarvisNotificationListenerService	System service capturing incoming notifications from WhatsApp, Telegram, SMS, and Gmail into a circular memory buffer for spoken readouts.
MemoryRepository	Persistent on-device key-value store managing user identity, custom names, and relationship contacts.
JarvisAccessibilityService	Performs system gestures: Screenshots, back, home, recents, notification shade, quick settings, power menu, split screen, and reel swiping.
CommandParser & ActionExecutor	Rule-based NLP engine using compiled RegEx and N-best candidate evaluation, routing intents to telephony, hardware, apps, and TTS.
JarvisTts	Cinematic British male butler speech profile (en-gb-x-rjs-local at pitch 0.90 / speed 0.98) with automated URL and symbol sanitization.

3. Complete Version 2.0 Command Reference

Category & Voice Command	Action Executed & System Behavior
--- Activation & Wake Up ---	
Hold Power Button	Silently activates bottom Siri pill with haptic tick.
Hey Jarvis / Jarvis	Wakes up Jarvis via background voice listener.
--- Spoken Notifications (v2) ---	
Read notifications / Read my notifications	Reads recent notifications from all apps.
Read messages / Read WhatsApp messages	Reads unread messages specifically from messaging apps.
Clear notifications	Clears internal notification readout history.
--- Context Memory & Relationships (v2) ---	
My name is [Name] / Call me [Name]	Stores user name in local memory.
What is my name / Who am I	Announces user's remembered name.
Remember [Contact] is my [Relation]	Maps contact (e.g. 'Remember Teja is my brother').
Who is my [Relation]	Queries relationship (e.g. 'Who is my brother?').
Clear memory / Forget everything	Erases all custom relationship and user memory.
--- System & Accessibility Actions (v2) ---	
Take a screenshot / Screenshot	Captures screen instantly via Accessibility.
Open notifications / Notification shade	Expands Android notifications shade.
Quick settings	Opens quick settings toggles panel.
Power menu	Displays device power / restart menu.
Split screen	Toggles split-screen multitasking.
--- Deep Media & Music (v2) ---	
Play [Song] on Spotify	Directly launches Spotify search and playback.
Play [Video/Song] on YouTube	Directly launches YouTube search and playback.
--- Conversational & Diagnostics (v2) ---	
Status report / System diagnostics	Tactical report: battery %, charging, service health.
Who are you / What are you	Speaks Jarvis assistant introduction.
How are you	Speaks system health check reply.
Tell me a joke	Delivers clean, witty British humor.
Thank you / Good job	Speaks courteous butler acknowledgement.
--- Routines & Modes ---	
Focus mode / Work mode	Silences ringers, enables DND, disables torch.
Night mode / Good night	Enables DND, mutes media, turns off torch.
Morning mode / Good morning	Disables DND, speaks time and battery status.
--- Phone Calls & WhatsApp ---	
Call [Name / Relation]	Calls contact or mapped relation (e.g. 'Call my brother').
Call [Name] on speaker	Places phone call with automatic speakerphone routing.
Call [Name] on WhatsApp	Initiates direct WhatsApp VoIP call.
Accept / Answer	Answers incoming call.
Decline / Reject	Rejects incoming call.

Silence call	Mutes incoming call ringer.
--- App & Browser Control ---	
Open [App Name]	Launches installed app with fuzzy match.
Open [Alias] (Insta, YT, WA, FB)	Launches app via shorthand nickname.
Open YouTube on Brave	Navigates browser directly to specified site.
Open the link I copied	Reads clipboard URL and launches in Chrome.
--- Gestures & Media Controls ---	
Go back / Go home / Recent apps	Standard Android system navigation gestures.
Next reel / Previous reel / Like reel	Hands-free short-form video feed navigation.
Scroll up / Scroll down	Smooth vertical screen scrolling.
Volume up / Volume down / Set volume [0-15]	Audio volume control.
Torch on / Torch off / Toggle flashlight	Device camera LED torch control.

4. Required Permissions for Version 2.0

To enable all v2 features on your phone, ensure the following permissions are granted:

- **Default Digital Assistant:** Settings → Apps → Default Apps → Digital Assistant App → Select JARVIS
- **Display Over Other Apps:** Settings → Apps → Special App Access → Display Over Other Apps → Enable JARVIS
- **Accessibility Service:** Settings → Accessibility → Downloaded Apps → JARVIS → Enable (Enables screenshots, gestures, navigation)
- **Notification Access:** Settings → Apps → Special App Access → Notification Access → Enable JARVIS (Enables spoken notification & message reader)
- **Microphone & Phone Permissions:** Record Audio, Contacts, Phone Calls, Phone State granted.
- **Battery Optimization:** Set to Unrestricted for uninterrupted background wake-word listening.